Visualizing Data with Graphs

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The Purpose of Data Visualization

"Graphs are not just decorative. They are tools for reasoning about data."

- **The Goal:** Represent data clearly to reveal distributions, differences, outliers, or trends over time.
- **The Challenge:** Moving from raw numbers to interpretation.
- **The Golden Rule:** The best graph matches the structure of the data and helps the reader answer a specific question.
- **Before graphing, ask:**
  1. What kind of variables am I trying to represent?
  2. What pattern do I want the reader to see?

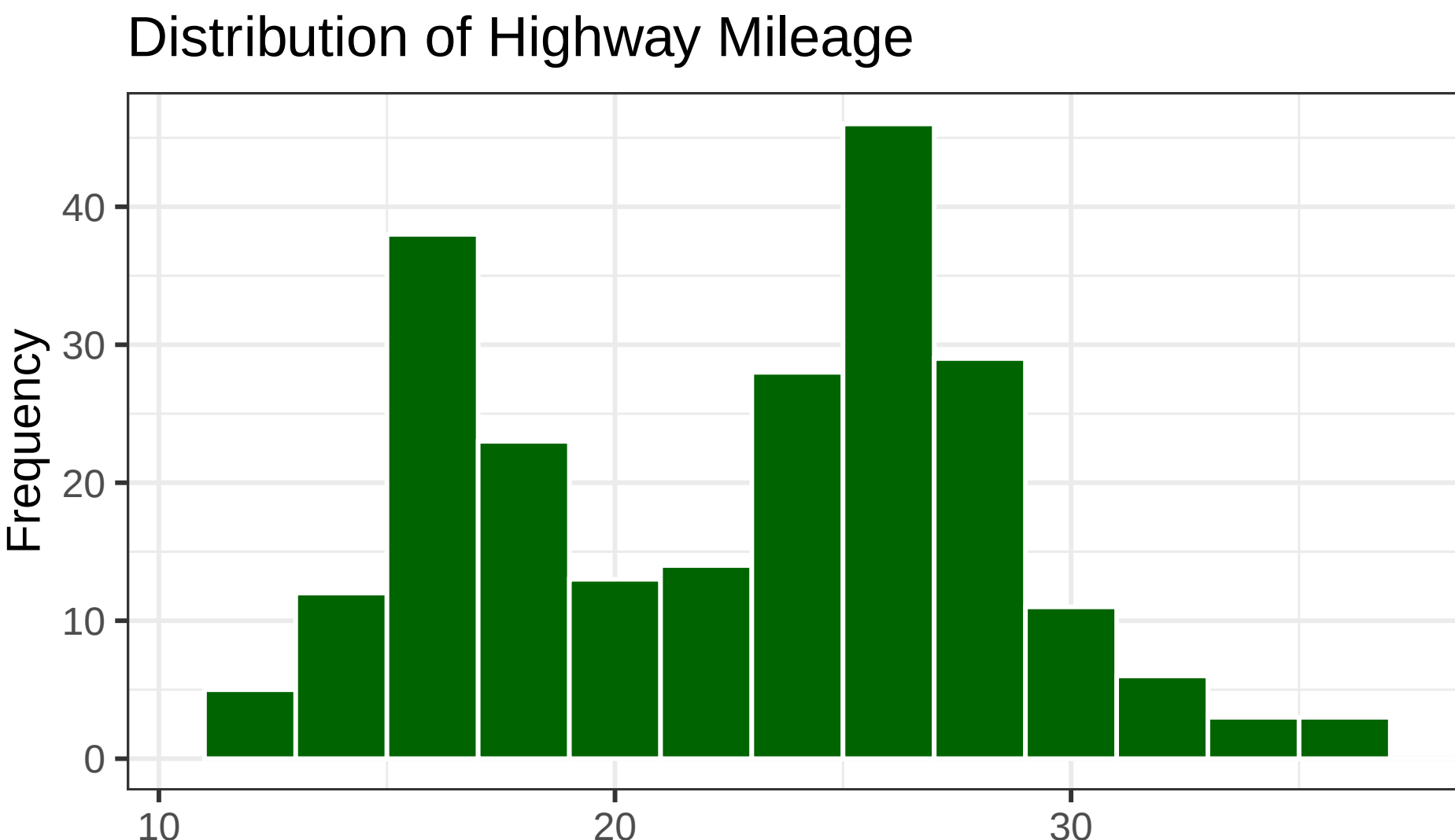
Choosing the Right Graph

Match your graph to your **data type** and **research question**:

- **Categorical Variables:**
  - **Pie Chart:** Parts of a whole (use sparingly).
  - **Column Chart:** Comparing counts, proportions, or descriptive statistics across categories.
- **Continuous Variables:**
  - **Histogram:** Distribution of one numeric variable.
  - **Boxplot:** Comparing distributions across groups.
  - **Line Chart:** Trends across ordered values (especially time).
  - **Scatterplot:** Relationships between two continuous variables.

Histograms

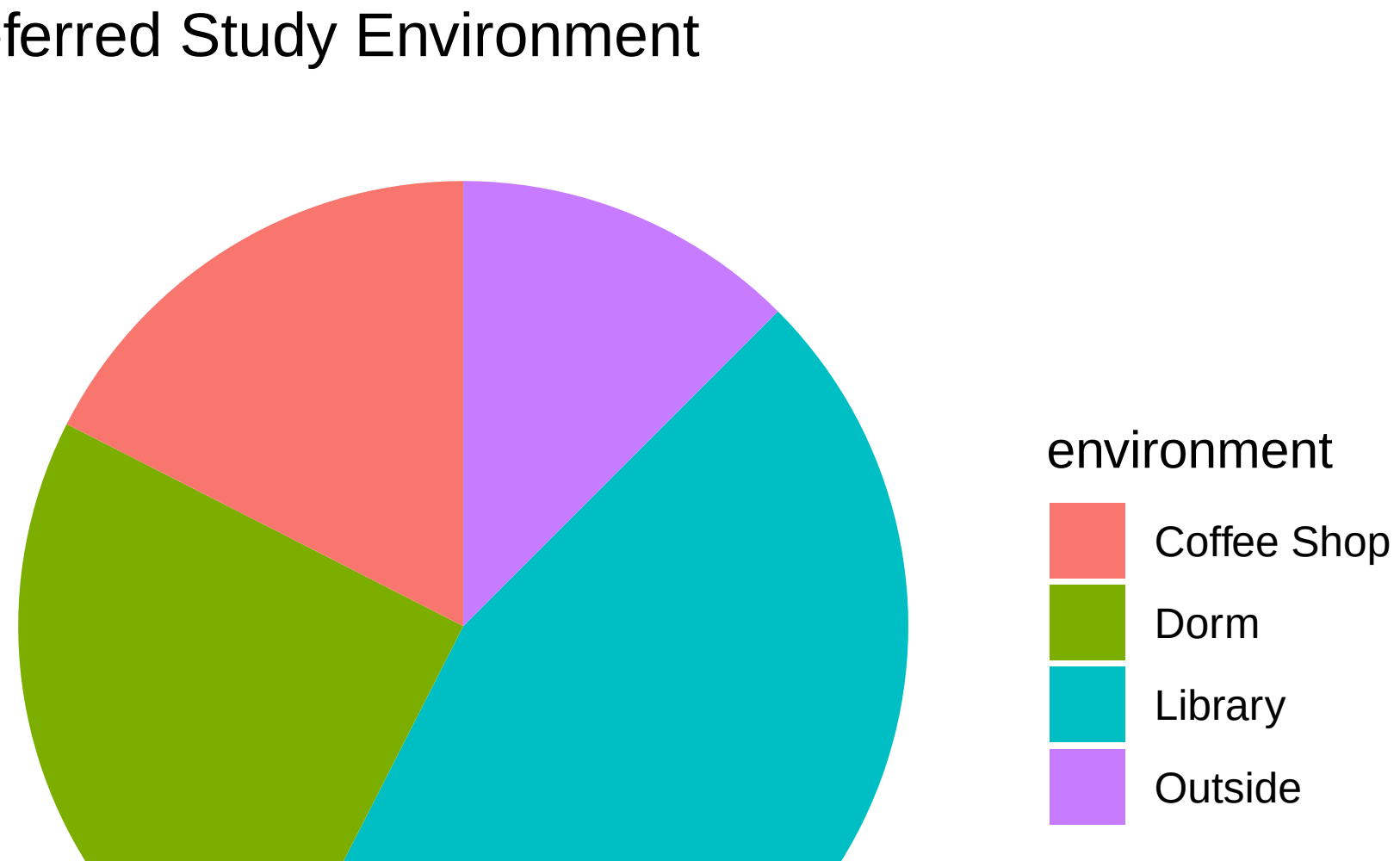
When to Use: - Examining the distribution of one continuous variable. - Checking for symmetry, skew, multimodality, or gaps.
When NOT to Use: - Categorical variables. - Comparing a large number of groups.



Pie Charts

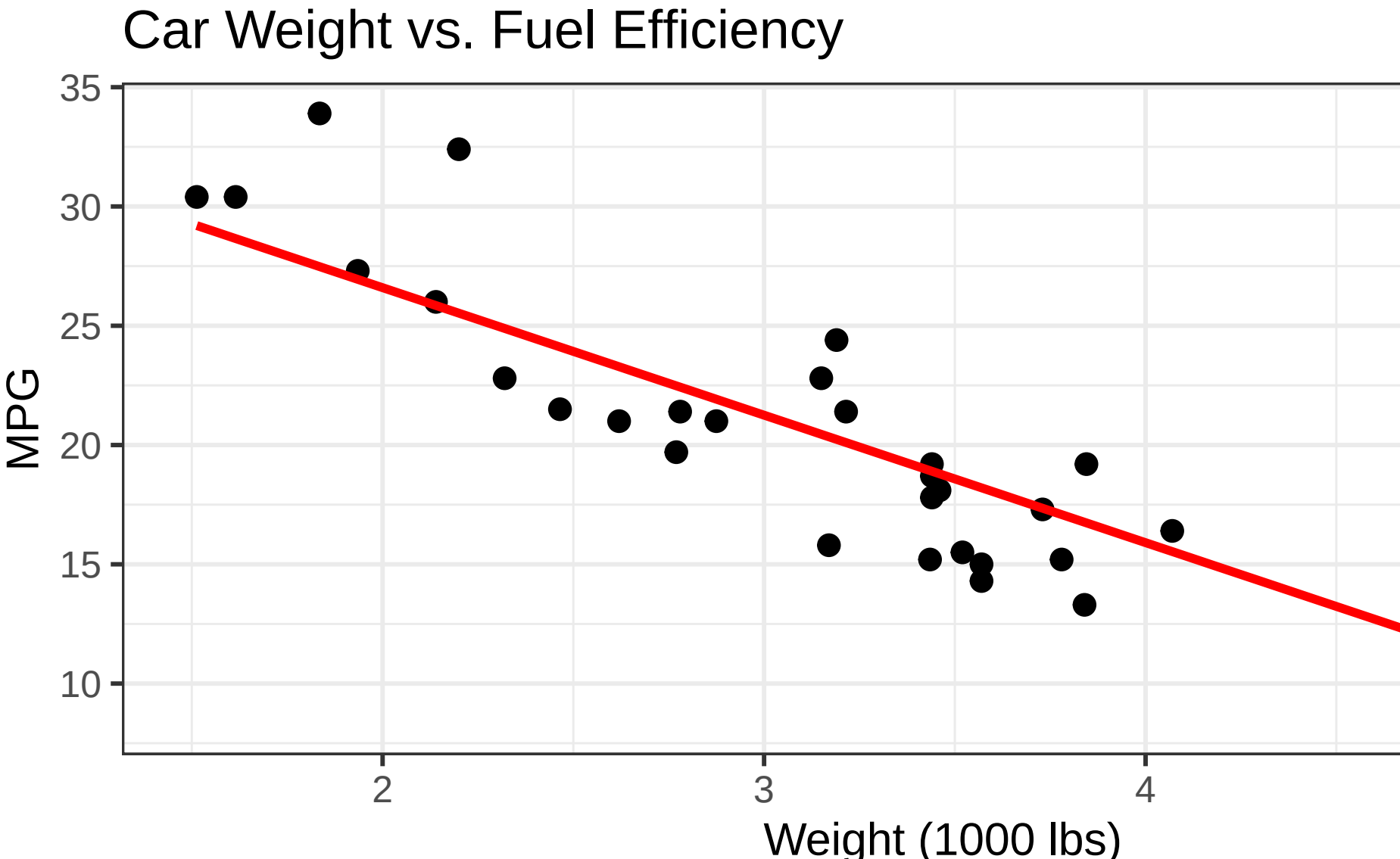
When to Use: - Categorical variables. - Small number of categories. - Goal is to emphasize relative proportions (parts of a whole).

When NOT to Use: - Many categories (becomes unreadable). - Categories have very similar proportions (angles are hard to compare).



Scatterplots

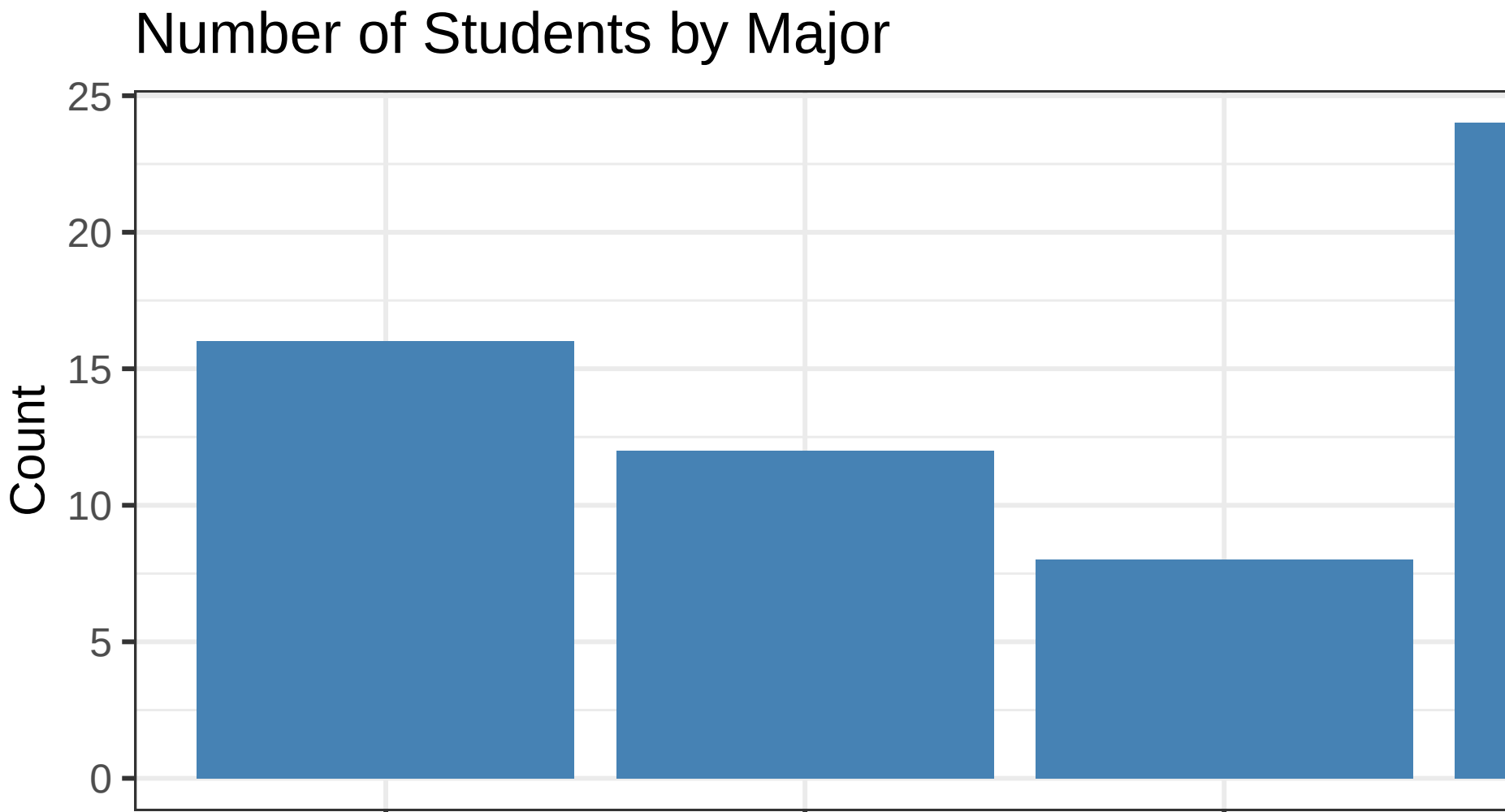
When to Use: - Both variables are quantitative/continuous. - Examining associations (direction, strength, and form).
When NOT to Use: - When variables are categorical.



Column Charts

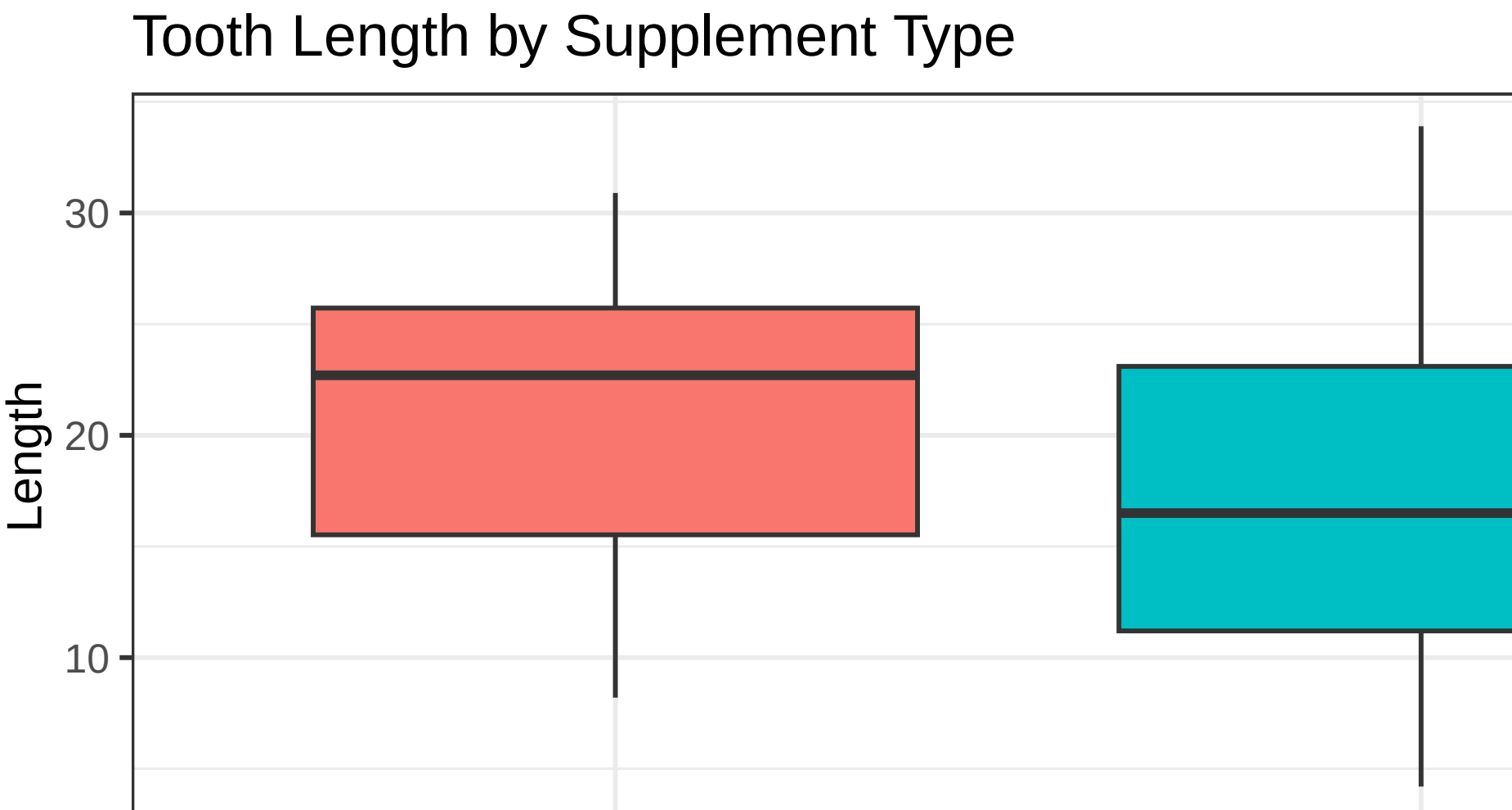
When to Use: - Comparing counts or means across discrete categories.

When NOT to Use: - Showing the shape of a continuous distribution. - Showing relationships between two continuous variables.



Boxplots

When to Use: - Comparing the distribution of a continuous variable across groups. - Showing center (median), spread (IQR), and outliers compactly.
When NOT to Use: - When the exact shape/detail of the distribution is necessary. - Very small sample sizes.



APA-Style Figures

When presenting results, figures must be clear, professional, and disciplinary-consistent.

Hallmarks of an APA-Style Figure: - Simple, readable fonts and clear axis labels. - No unnecessary background effects or 3D decoration. - Minimal visual clutter (prefer `theme_classic()`). - Informative captions explaining the variables and error bars.

> ***Rule of Thumb:*** *Prefer clarity over visual flair. If it prints well in grayscale, it's usually a good start.*

Error Bars: Showing Uncertainty

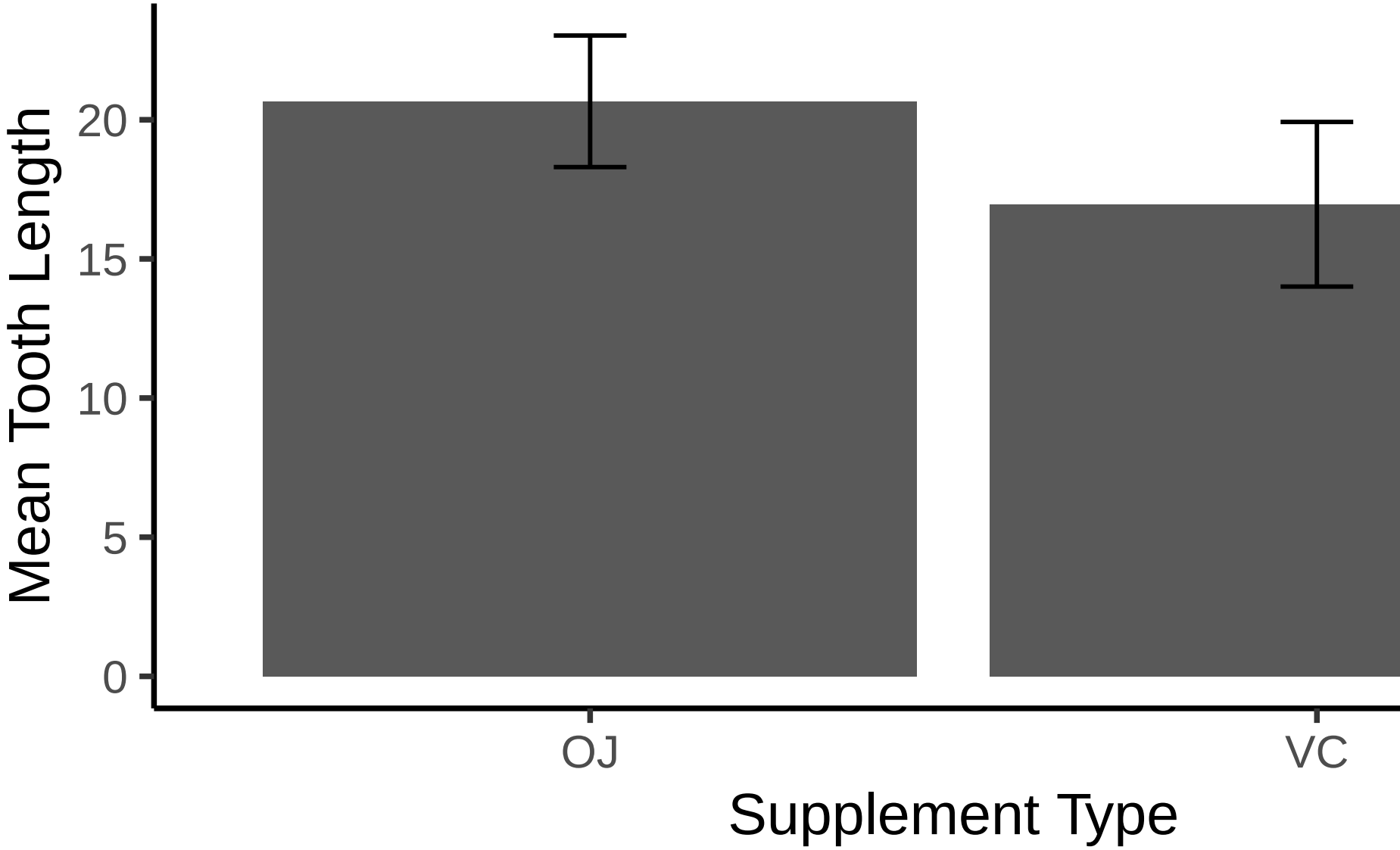
When graphing a group mean, the mean is only a summary. **Error bars** represent uncertainty or variability.

Types of Error Bars: 1. **Standard Deviation (SD):** Shows how spread out individual observations are. 2. **Standard Error (SE):** Shows how precisely the sample mean estimates the population mean. 3. **Confidence Interval (CI):** Gives a range of plausible values for the population parameter.

Warning: Always explicitly state what your error bars represent in your figure caption!

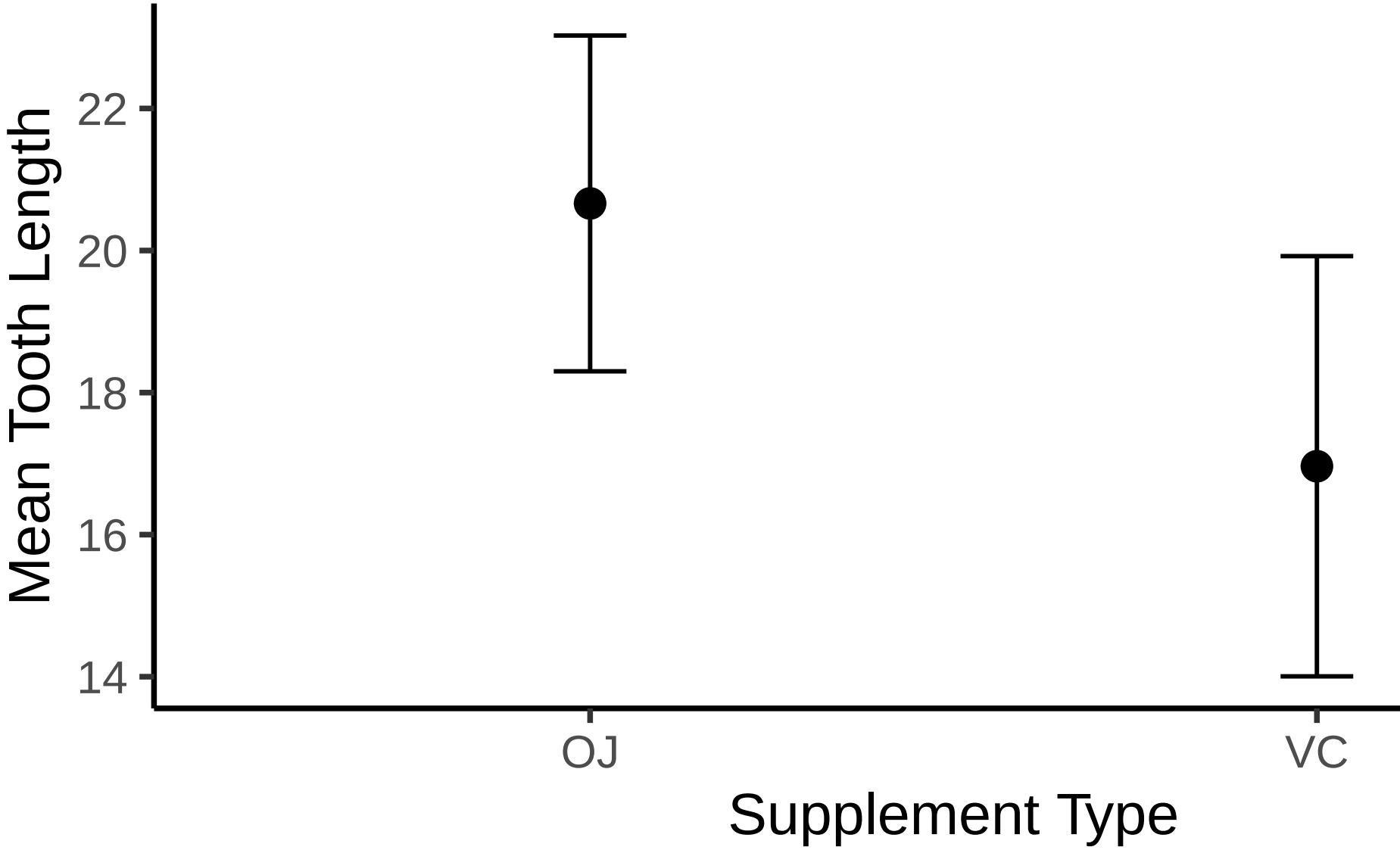
Error Bars - Column Chart

Point plots are often preferred over bar charts for showing means, as they emphasize the estimate rather than the raw total.



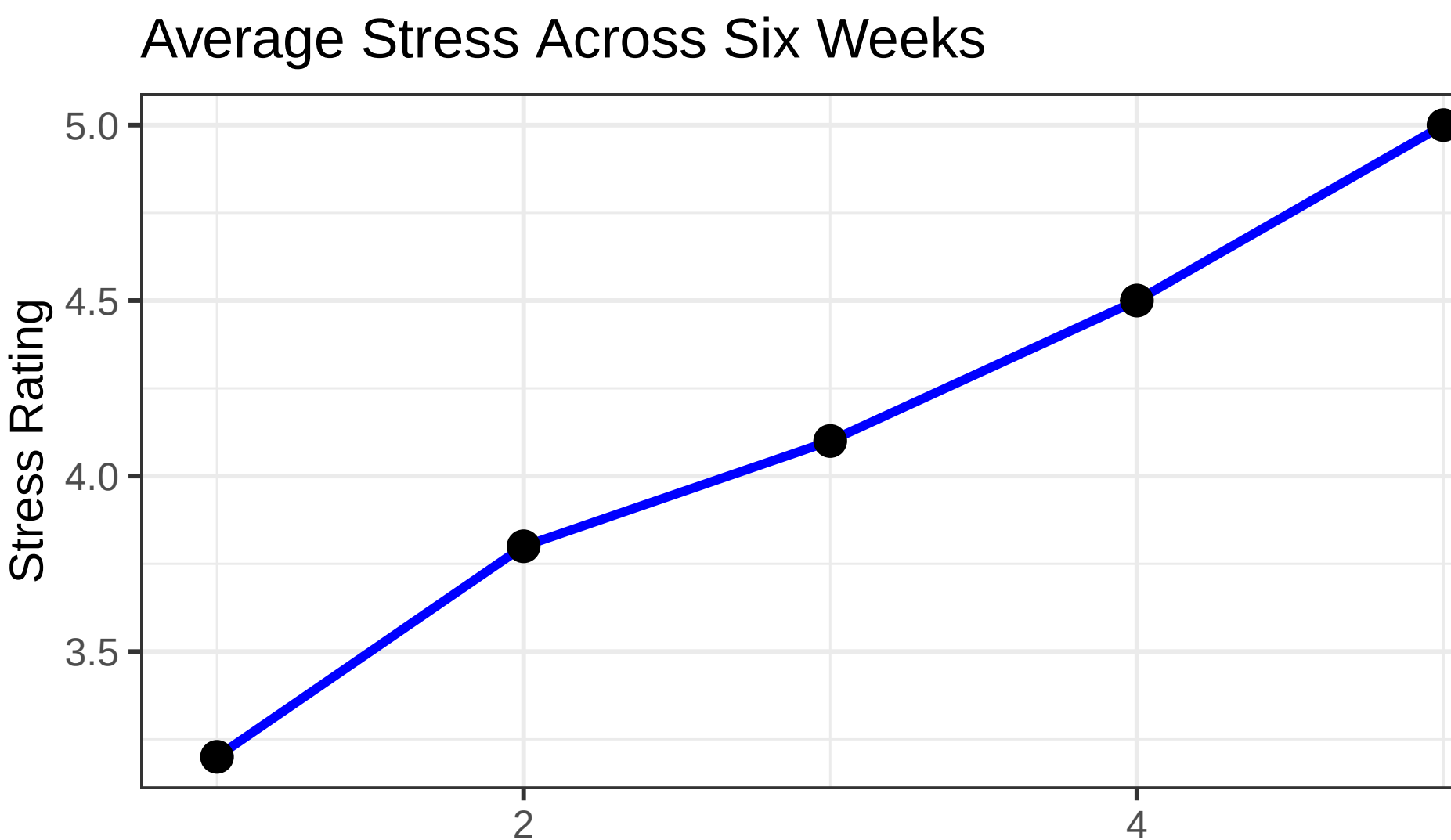
Error Bars - Point Plot

Point plots are often preferred over bar charts for showing means, as they emphasize the estimate rather than the raw total.



Line Charts

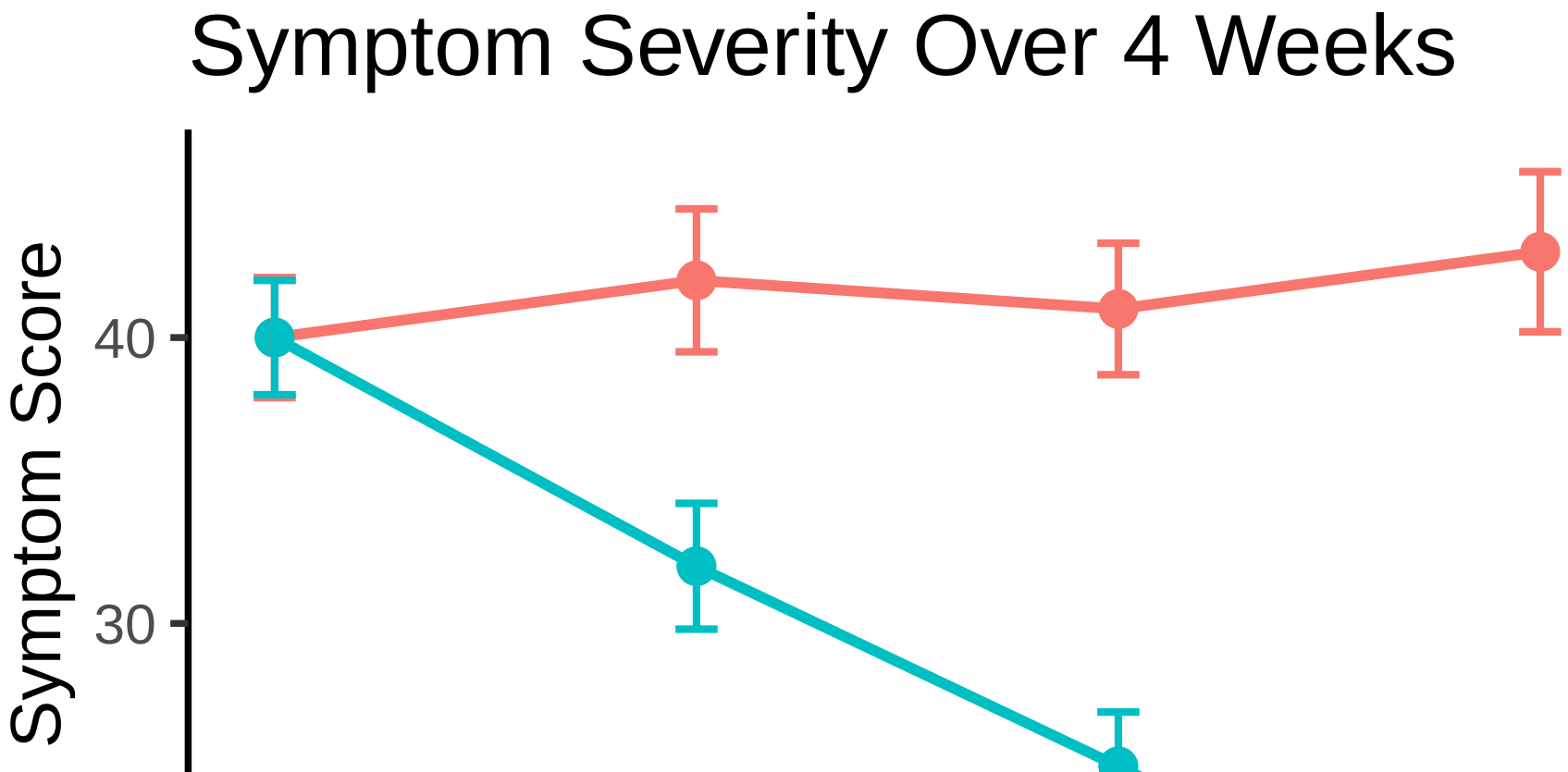
When to Use: - X-axis is logically ordered (e.g., Time). - Highlighting trends, trajectories, or repeated measurements.
When NOT to Use: - Unordered categories (e.g., comparing Biology to Psychology).



Line Charts: Multiple Groups & Error Bars

When to Use: - You have an ordered sequence (like time) on the x-axis. - You are comparing **two or more groups** across that sequence. - You need to show both the trend (the line) and the uncertainty (the error bars).

Crucial R Tip: To draw separate lines for each group, you must tell `ggplot2` to group the data using `group = your_category_variable` inside the `aes()` function!



Common Mistakes to Avoid

1. **Pie chart overload:** Using a pie chart for too many categories hides small differences.
2. **Categorical continuous data:** Using a column chart for a continuous distribution (use a histogram instead).
3. **False continuity:** Using a line chart for unordered categories.
4. **Hiding the spread:** Using a chart of means when variability/skew heavily impacts the interpretation (use a boxplot instead).
5. **Chart junk:** Adding 3D effects, heavy gridlines, or excessive colors that distract from the data.

Final Takeaway

> *"Plots and graphs are not just decorative. They are tools for reasoning about data."*

Ask yourself **"What relationship, pattern, or comparison am I trying to show?"** That question will usually point you to the correct graph.

Making Graphs in Jamovi

Many plots/graphs can be generated directly from the Jamovi menus and/or within the options of the analysis you are using.

When Jamovi lacks the option to make a plot/graph, you can use the RjEditor plugin.

Scatterplots in Jamovi

1. Identify two continuous variables
2. Select `Explore > Scatterplot`
3. Add your variables to the X and Y axis fields
4. Select options as desired

Boxplots/Bar plots/Histograms in Jamovi

1. Identify a continuous variable and a categorical (grouping) variable
2. Select `Explore > Descriptives`
3. Add your continuous variable to the "Variables" field
4. Add your categorical variable to the "Split-by" field
5. Select options as desired

Grouped Boxplots/Bar plots in Jamovi

1. You can create grouped bar charts and boxplots by adding more categorical variables to the “Split-by” field

Exporting Figures

1. right-click > image > export

Choose .png as file type when saving

Cleaning Up Jamovi Figures

Editing Figures

- Jamovi Graphs/Plots can be easily manipulated in many different programs
- I use Google Slides!

Clean Figure